and largest underwater robot used in scientific exploration projects such as exploring shipwrecks as well as fixing structures like the pipes used to carry oil and gas far beneath the waves.

Transforming aqua-culture and marine management

Marine species are important for scientific researches and commercial uses. Many researchers and agencies have been deploying AI to prepare a specific database of marine species for marine management.

Recently, the AI platform developed in association with Microsoft is an attempt to automate the exhaustive process of counting local fish stocks and identify the different varieties of fish in the oceans. Machine learning is often applied to areas where there is a lot of uncertainty and unknown information is to be collected and processed. Researchers have integrated cloud computing with AI to come up with solutions specific to their marine applications. They claim that their machine learning model is now capable of identifying the species with high accuracy.

An AI program developed by the researchers at North Carolina State University can automatically identify species or microscopic marine organisms. This could be the first step in developing a robotic system that can provide a clearer picture of the world's oceans, both past and present. The program is capable of identifying six species of foraminifera (or forams). These organisms have been prevalent in the Earth's oceans for more than 100 million years. By examining shells, the scientists can determine when the shells were formed. Scanning and identification take only seconds, and correctly identify the forams faster than human experts.

Mapping the oceans

AI with sensors are capable of recording vast amounts of data about the environment, including monitoring temperature, marine life, tsunami, earthquake, etc. Underwater drones and portable submersible robots with intelligent sensors for monitoring water temperature, auto compass heading and depth, turns, and pitch-and-roll have also been developed.

Fig. 3: Artificial island in South China Sea (Credit: www.scmp.com)

The next-generation ocean monitoring system's development has already begun in the US, Japan and European countries. The US Ocean Observatories Initiative has been providing researchers with spatial and temporal data from the oceans' surface all the way down to the deep-sea. The initiative is planning to launch robots to chart the world's oceans, with the goal of mapping everything by 2030.

First underwater AI colony

So far, ocean colonisation is just a theory of human settlement in the oceans. The purpose of ocean colonisation is the expansion of liveable area, access to undersea resources, new recreational activities and others. Human settlement in the oceans may take some years, but robots and AI machines could be deployed to make base stations in the oceans, like space stations in space.

Chinese scientists are set to create the world's first AI colony on South China Sea, under the project known as Deep Sea Atlantis. According to sources, researchers will construct the base between 6km (19,685-feet) and 11km (36,100-feet) below the South China Sea's surface. Exact location of the project is not known, but report suggests that the base will be set up near the artificial island in South China Sea.

The base will have docking platforms like a space station. Robotic submarines will leave the base from these stations to conduct exploratory missions, surveying new areas and collecting data about marine life forms.

As per sources, Chinese President Xi Jinping told media that the AI-based colony will be used for defence operations and unmanned submarine science.

To sum up

An in-depth insight into the mysteries of the oceans has been obtained from the analysis of deep-sea images using AI. A standard workflow helps oceanographers to systematically evaluate large quantities of image data being captured from the oceans.

AI and autonomous robots are used to speed up the exploration and study of hard-to-reach deep-sea ecosystems. Improvements in AI and robotics in ocean exploration are likely to benefit researchers in multiple areas, especially those handling dangerous tasks. Several giants in the oil and gas industries are integrating AI for innovation and sustainable energy strategies, driving the pace of evolution in the field. Countries like China are planning for an AI colony in the ocean to fully explore the resources for its defence applications and unmanned submarine science. EFY